

Modeling of the Cart and Track System

Figure 1 shows a cart on a track. The smaller wheel is powered by a DC motor. The larger wheel is connected to an encoder that gives the position of the cart.

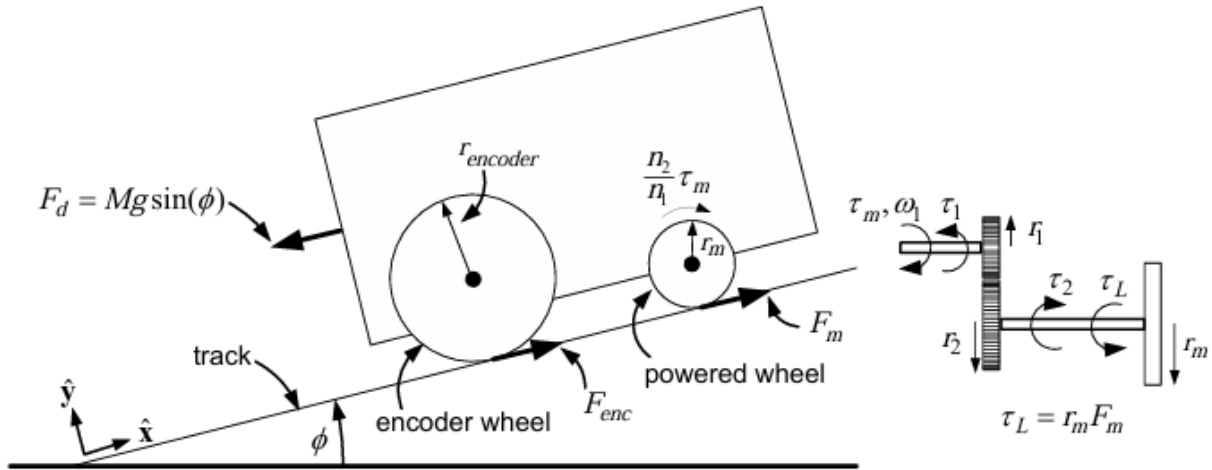


Figure 1: Cart on a track with one end raised an angle ϕ with respect to the horizontal.

A model is now developed for the cart system. The following notation is used.

K_T is the motor torque constant ($\tau = K_T i$)

$K_b (= K_T)$ is the motor back-emf constant ($v_b = K_b \omega_1$)

R is the motor armature resistance

L is the motor armature inductance

$M = M_c$ is the mass of the cart (without the extra weight mass M_w)

r_m is the radius of the powered (motor) wheel

r_1 is the radius of the gear on the motor shaft

r_2 is the radius of the gear on the wheel shaft

J_1 is the moment of inertia of the motor shaft

J_2 is the moment of inertia of the output shaft and powered wheels

r_{enc} is the radius of the encoder wheel

J_{enc} is the moment of inertia of the encoder shaft and wheels.

f_1 is the viscous friction coefficient of the motor shaft

f_2 is the viscous friction coefficient of the output shaft

f_{enc} is the viscous friction coefficient of the encoder shaft

n_1 is the number of gear teeth on the motor shaft

n_2 is the number of gear teeth on the output shaft

$r_g = n_2/n_1 = r_2/r_1$ is the gear ratio¹

$F_d = Mg \sin(\phi)$ is the (disturbance) force of gravity along the direction of the track.

Mechanical Equations To model the mechanical part of the system, both the rotational motion of the motor, gears and wheels along with the translational motion of cart must be taken into account.

Denote by $F_m \hat{\mathbf{x}}$ the force exerted on the powered wheels by the track so that (Newton's third law) $-F_m \hat{\mathbf{x}}$ is the force exerted on the track by the powered wheels. Similarly, let $F_{enc} \hat{\mathbf{x}}$ be the force exerted on the encoder wheels by the track so that $-F_{enc} \hat{\mathbf{x}}$ is the force exerted on the track by the encoder wheels. Gear 1 is on the motor shaft while gear 2 is on the wheel shaft. For the motorized wheels, we then have

$$J_1 \frac{d\omega_1}{dt} = \tau_m - \tau_1 - f_1 \omega_1 \quad (1)$$

$$J_2 \frac{d\omega_2}{dt} = \tau_2 - r_m F_m - f_2 \omega_2 \quad (2)$$

where τ_1 is the torque exerted on gear 1 by gear 2 and τ_2 is the (reaction) torque exerted on gear 2 by gear 1. The quantity $r_m F_m$ is the (load) torque exerted on the powered wheel by the track. Using the gear relationships

$$\omega_1 = \frac{n_2}{n_1} \omega_2, \tau_2 = \frac{n_2}{n_1} \tau_1$$

the two equations (1) and (2) reduce to the single equation

$$\underbrace{\left(J_1 \left(\frac{n_2}{n_1} \right)^2 + J_2 \right)}_J \frac{d\omega_2}{dt} = \frac{n_2}{n_1} \tau_m - F_m r_m - \underbrace{\left(f_1 \left(\frac{n_2}{n_1} \right)^2 + f_2 \right)}_f \omega_2 \quad (3)$$

or

$$J \frac{d\omega_2}{dt} = \frac{n_2}{n_1} \tau_m - F_m r_m - f \omega_2. \quad (4)$$

The encoder wheels have a moment of inertia J_{enc} , radius r_{enc} and angular velocity ω_{enc} . The equation of motion for the encoder wheel is then

$$J_{enc} \frac{d\omega_{enc}}{dt} = -F_{enc} r_{enc} - f_{enc} \omega_{enc} \quad (5)$$

where the quantity $r_{enc} F_{enc}$ is the torque exerted on the encoder wheels by the track.

The linear speed along the circumference of the encoder wheels is $r_{enc} \omega_{enc}$ and is the same as the linear speed along the circumference of the powered wheels $r_m \omega_2$ as they make contact with the inclined plane and we are assuming there is no slip. This common linear speed is simply the speed of the cart $v = dx/dt$. This is all written mathematically as

$$r_{enc} \omega_{enc} = r_m \omega_2 = v. \quad (6)$$

The equation of motion for the cart is simply

$$M \frac{dv}{dt} = F_m + F_{enc} - F_d. \quad (7)$$

It turns out that $J_{enc}, f_1, f_2, f_{enc}$ are all negligible so we take them to be zero. The equation for the encoder wheel given by (5) simply goes away as $f_{enc} \approx 0, J_{enc} \approx 0$ so that $F_{enc} \approx 0$. The equations of mechanical motion (4) and (7) are then

$$J \frac{d\omega_2}{dt} = \frac{n_2}{n_1} \tau_m - F_m r_m \quad (8)$$

$$M \frac{dv}{dt} = F_m - F_d. \quad (9)$$

Using the no slip condition (6), i.e., $\omega_2 = v/r_m$, for the powered wheel and eliminating F_m from (8) and (9), we obtain the single equation of motion for the speed v of the cart given by

$$\left(\frac{J}{r_m} + r_m M \right) \frac{dv}{dt} = \frac{n_2}{n_1} \tau_m - r_m F_d$$

or equivalently

$$\left(\frac{J}{r_m^2} + M \right) \frac{dv}{dt} = \frac{1}{r_m} \frac{n_2}{n_1} \tau_m - F_d.$$

Electrical Equations The motor electrical equations are given by (see Figure 2)

$$L \frac{di}{dt} = -Ri(t) - K_b \omega_1(t) + v_a(t)$$

and the torque produced by the motor is

$$\tau_m = K_T i(t).$$

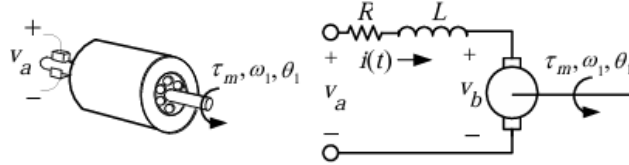


Figure 2: Schematic diagram of a DC motor. $v_b = K_b \omega_1 = K_b \frac{n_2}{n_1} \frac{v}{r_m}$, $\tau_m = K_T i$.

By the gear relationship and assuming no slip between the cart wheel and the track, we have

$$\omega_1 = \frac{n_2}{n_1} \omega_2 = \frac{n_2}{n_1} \frac{v}{r_m}$$

so that the electrical equation becomes

$$L \frac{di}{dt} = -Ri(t) - \underbrace{\frac{K_b n_2}{r_m n_1} v(t)}_{v_b} + v_a(t).$$

Equations of Motion and Block Diagram Collecting the above equations together, the cart system is described by²

$$\begin{aligned} L \frac{di}{dt} &= -Ri(t) - \underbrace{\frac{K_b n_2}{r_m n_1} v(t)}_{v_b} + v_a(t) \\ \left(\frac{J}{r_m^2} + M \right) \frac{dv}{dt} &= \left(\frac{n_2}{n_1} \frac{1}{r_m} \right) K_T i(t) - F_d \\ \frac{dx}{dt} &= v. \end{aligned} \tag{10}$$

It is easier to work with these equations in the s domain. Taking the Laplace transform of the equations of system (10) gives

$$\begin{aligned} I(s) &= \frac{V_a(s) - V_b(s)}{sL + R} \\ V_b(s) &= \underbrace{K_b \frac{n_2}{n_1} \frac{1}{r_m}}_{K'_b} v(s) \\ \underbrace{s \left(\frac{J}{r_m^2} + M \right)}_{M'} v(s) &= \underbrace{\frac{n_2}{n_1} \frac{1}{r_m} K_T}_{K'_T} I(s) - F_d(s). \end{aligned}$$

where we simplify the notation by letting

$$K'_T \triangleq K_T \frac{n_2}{n_1} \frac{1}{r_m}, \quad K'_b \triangleq K_b \frac{n_2}{n_1} \frac{1}{r_m}, \quad M' \triangleq J/r_m^2 + M.$$

A block diagram representation of these relationships is given in Figure 3.

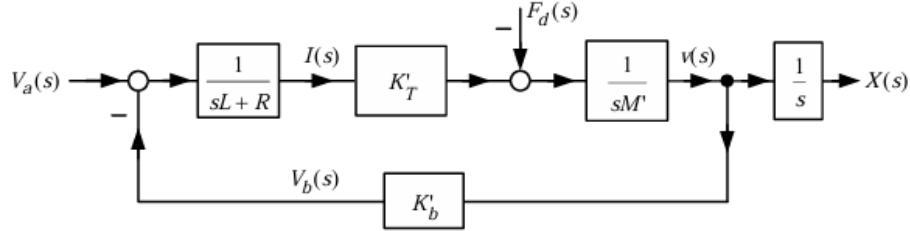


Figure 3: Block diagram of the cart on a track.

Setting $L = 0$ (as it is negligible) and moving the disturbance force (due to gravity) to the input summing junction, the block diagram of Figure 3 reduces to that of Figure 4 where

$$K_D \triangleq \frac{1}{K'_T/R} = \frac{R}{K'_T \frac{n_2}{n_1} \frac{1}{r_m}}.$$

²Remember that $v(t) = dx/dt$ is the linear speed of the cart!

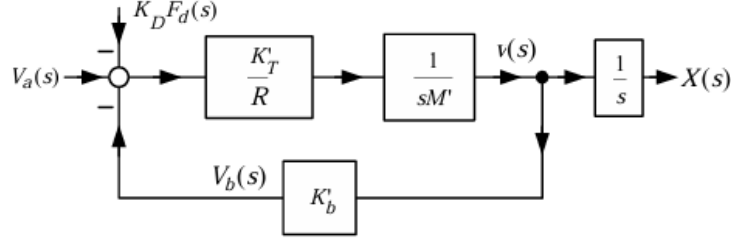


Figure 4: Equivalent block diagram for the cart and track system.

We can simplify this block diagram further by defining $G(s)$ as

$$G(s) \triangleq \frac{\frac{K'_T}{R} \frac{1}{sM'}}{1 + K'_b \frac{K'_T}{R} \frac{1}{sM'}} \frac{1}{s} = \frac{K'_T}{sRM' + K'_b K'_T} \frac{1}{s} = \frac{\frac{K'_T}{RM'}}{s + \frac{K'_b K'_T}{RM'}} \frac{1}{s} = \frac{b}{s(s+a)}$$

where

$$a \triangleq \frac{K'_b K'_T}{RM'} = \frac{K_b K_T \left(\frac{n_2}{n_1} \frac{1}{r_m} \right)^2}{RJ/r_m^2 + RM}, \quad b \triangleq \frac{K'_T}{RM'} = \frac{K_T \frac{n_2}{n_1} \frac{1}{r_m}}{RJ/r_m^2 + RM}, \quad bK_D = \frac{1}{J/r_m^2 + M}.$$

We see from Figure 4 that we then have

$$X(s) = \frac{b}{s(s+a)} V_a(s) - \frac{b}{s(s+a)} K_D F_d(s)$$

so that our block diagram reduces to that given in Figure 5 below.

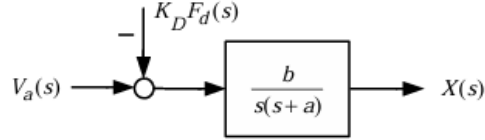


Figure 5: Equivalent transfer function model of the cart and track system.

Optical Encoder Model

A common position sensor used in industry is an optical encoder which is illustrated in Figure 6(a) below [1]. As shown, the optical encoder consists of a set of windows spaced equally around a circular disk with a light source shining through the window when it is aligned with the source. A detector puts out a high voltage when there is light and a low voltage otherwise. For the setup of Figure 6(a), there are 12 windows (lines or slots) so that for every complete revolution of the

circular disk (i.e., of the motor), there will be 12 pulses. Using digital electronic circuitry, one can detect a pulse going high or low so that in one revolution with 12 pulses, there will be a total of 24 times that a pulse went either high or low. Note that each time a pulse goes high or low, the motor has rotated $2\pi/24$ radians or $360^\circ/24 = 15^\circ$. By simply counting the number N of rising and falling edges of the pulse, one can obtain the position of the rotor to within 15° .

In order to detect the direction of rotation, two light detectors are used as shown in Figure 6(b). In detail, the length of the windows is the same as the length of the distance between windows. The two light detectors are placed a distance apart equal to $1/2$ of a window length. One period of the voltage waveform coming out of the light detector corresponds to the distance from the beginning of one window to the next and this cycle of the voltage waveform is considered to be 360° as illustrated in Figure 6(b). Consequently, the two light detectors are considered to be 90° apart or equivalently, they are said to be in *quadrature*.

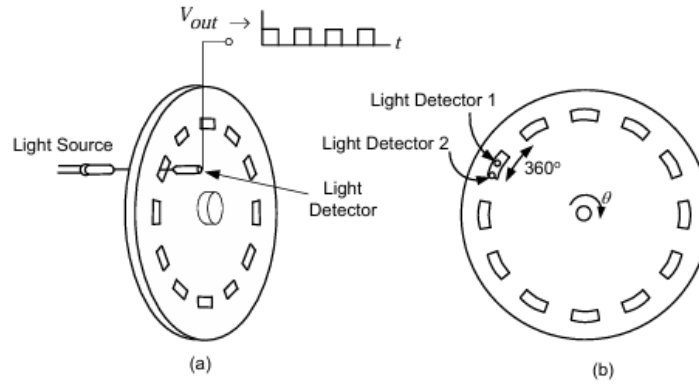


Figure 6: Schematic diagram of an optical encoder [1].

Figure 7(a) shows the voltage waveforms out of the two light detectors when the rotor is turning *clockwise*.

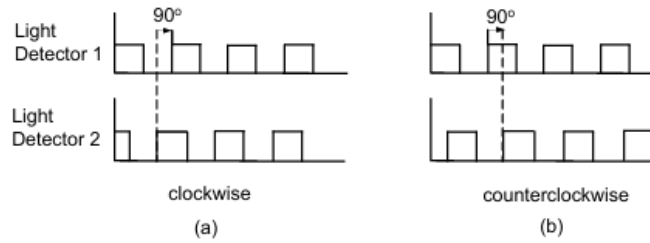


Figure 7:

As shown, in the case of clockwise rotation, the voltage of light detector 1 is 90° behind that of light detector 2. That is, the voltage from light detector 2 goes high first and then a quarter of a cycle

later, the voltage from light detector 1 goes high. However, if the rotor is turning *counterclockwise*, then the voltage of light detector 2 is 90° behind that of light detector 1 as shown in Figure 7(b). Consequently, the encoder electronics detects the relative phase of the two light detector voltage signals and uses that to determine whether the rising or falling edges of a pulse should increase the count (clockwise motion) or decrease the count (counterclockwise motion). If an optical encoder has N_w windows (lines/slots), then there are $2N_w$ rising and falling edges per revolution giving a resolution of $2\pi/(2N_w)$ radians. If one counts the voltage pulses from *both* light detectors, there will be $4N_w$ rising and falling edges per revolution so that a resolution of $2\pi/(4N_w)$ radians is achieved. For example, with $N_w = 500$, the resolution of the encoder is $2\pi/2000$ radians or $360^\circ/2000 = 0.18^\circ$.

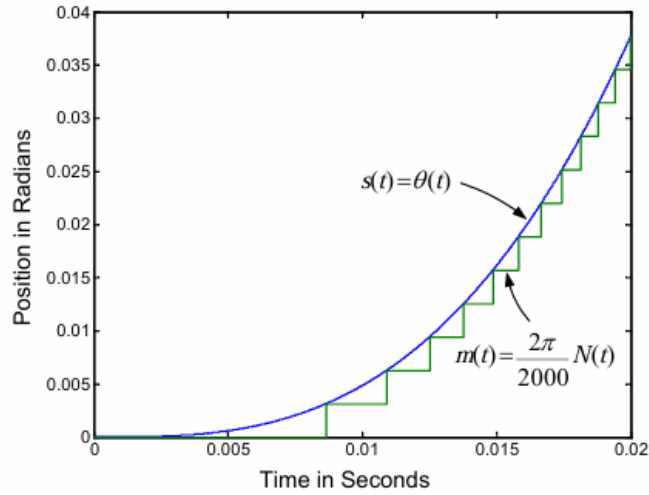


Figure 8: Plot of $\theta(t)$ and the encoder output $(2\pi/2000)N(t)$.

Encoder model As above r_{enc} denotes the radius of the encoder wheel and let N_{enc} denote the number of counts per revolution put out by the encoder. Then

$$K_{enc} = \frac{2\pi}{N_{enc}} r_{enc} \text{ meters/count}$$

is the conversion from encoder counts to linear position.

Backward Difference Estimation of Speed The optical encoder gives the position measurement, but not the speed of the cart. However, one can use this measurement to deduce the speed. The most straightforward way is to compute the *backward difference* of the position and divide by the sample period, i.e.,

$$\hat{v}_{bd}(kT) \triangleq K_{enc} \left(\frac{N(kT) - N(kT - T)}{T} \right)$$

where T is the time between samples and $N(kT)$ is the optical encoder count at time kT .

The error in estimating the speed by differentiation of the position measurements can be found as follows: At any discrete time kT , $N(kT)$ is in error by at most one encoder count. In particular, $N(kT)$ can be only too small by at most one encoder count ($N(kT)$ is never too large because of the way the encoder works as shown in Figure 8. Thus, with $x(kT)$ the true position of the cart in meters, we have

$$x(kT) = K_{enc}N(kT) + K_{enc}e(kT)$$

where $e(kT)$ represents the *positive fractional* count that the encoder cannot sense, that is, $0 \leq e(kT) < 1$ for all k . Then, the speed may be written as

$$\begin{aligned} v(kT) &= \left(\frac{x(kT) - x(kT - T)}{T} \right) \\ &= K_{enc} \left(\frac{N(kT) - N(kT - T)}{T} \right) + K_{enc} \left(\frac{e(kT) - e(kT - T)}{T} \right). \end{aligned}$$

As $0 \leq e(kT) \leq 1$ and $0 \leq e((k-1)T) \leq 1$, it follows that

$$|e(kT) - e(kT - T)| \leq 1.$$

It is now straightforward to compute a bound on the error in estimating the speed. As the speed estimate is given by

$$\hat{v}_{bd}(kT) = K_{enc} \left(\frac{N(kT) - N(kT - T)}{T} \right)$$

and the difference $e(kT) - e(kT - T)$ is bounded by ± 1 , it follows that

$$|\text{Error in } \hat{v}_{bd}(kT)| = K_{enc} \left| \frac{e(kT) - e(kT - T)}{T} \right| \leq \frac{K_{enc}}{T}.$$

As the sampling rate increases (T gets smaller), the error gets larger. On the other hand, as the sampling rate decreases this encoder resolution error becomes smaller, but the backward difference approximation

$$v(kT) \approx \left(\frac{x(kT) - x(kT - T)}{T} \right)$$

becomes less valid. The choice of T is a trade off between the encoder resolution error and the accuracy of the finite difference in approximating the derivative. One way to decrease this error would be to use an encoder with higher resolution. Such encoders are typically more expensive and cannot operate at higher speeds (as the speed increases, a large number of pulses are coming in so fast that the pulse detection circuitry cannot keep up).

System Setup and System Parameters

System Setup The system setup for cart on the track is illustrated in Figure 9. The desired value of voltage to the motor is developed on the computer and sent to the D/A (digital to analog converter) on the interface board. From the D/A the desired voltage value is input to the amplifier which produces the desired voltage on the motor. The encoder signal (a sequence of rising and falling voltages as in Figure 7) is read into the interface board which converts it to counts (integer valued) which is then input to the computer.

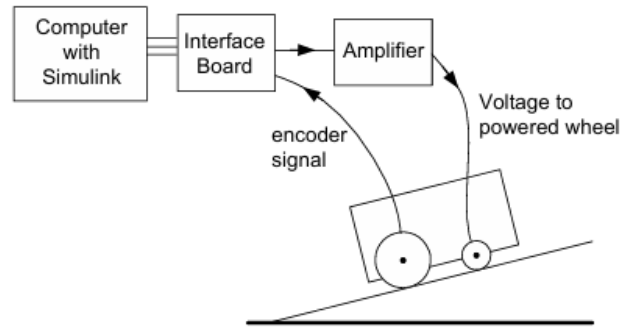


Figure 9: System Setup.

Parameter Values The following table contains the values of the parameters for the cart and track system as given by the manufacturer QUANSER.

Parameter	Value
K_T	$7.67 \times 10^{-3} \text{ Nm/A}$
R	2.6 Ohms
L	0.18 mH
M_c	0.57 kg
M_w	0.37 kg
r_m	$6.35 \times 10^{-3} \text{ m}$
J_1	$3.90 \times 10^{-7} \text{ kgm}^2$
J_2	$\approx 0 \text{ kgm}^2$
J_{enc}	$\approx 0 \text{ kgm}^2$
f_1	$\approx 0 \text{ Nm/rad/sec}$
f_2	$\approx 0 \text{ Nm/rad/sec}$
f_{enc}	$\approx 0 \text{ Nm/rad/sec}$
$r_g = n_2/n_1$	3.71 (dimensionless)
g	9.8 m/sec^2
N_{enc}	$4 \times 1024 = 4096 \text{ counts/rev}$
$K_{enc} = (2\pi/N_{enc})r_{enc}$	$2.275 \times 10^{-5} \text{ meters/count}$
V_{max}	5 V
T	$0.001 \text{ sec (sample period)}$

Parameter Identification In the transfer function model given in Figure 5, the only parameters that need to be identified (estimated) are a and b . This can be done using an experimental plot of speed versus time. To show how this is accomplished, consider the transfer function from input voltage $v_a(t)$ to output velocity $v(t) = dx/dt$ from Figure 5 given by ($\dot{X}(s) \triangleq \mathcal{L}\{dx/dt\}$)

$$\dot{X}(s) = \frac{b}{s+a} V_a(s).$$

With a step input voltage $v_a(t) = V_0 u_s(t)$ or

$$V_a(s) = \frac{V_0}{s}$$

we have

$$\dot{X}(s) = \frac{b}{s+a} \frac{V_0}{s} = V_0 \frac{b}{a} \left(\frac{1}{s} - \frac{1}{s+a} \right).$$

Computing the inverse Laplace transform gives the output velocity as

$$\frac{dx}{dt} = v(t) = V_0 \frac{b}{a} (1 - e^{-at}).$$

Making a table of values of $1 - e^{-at}$ versus time t results in

t	$1 - e^{-at}$
0	$1 - e^{-0} = 0$
$1/a$	$1 - e^{-1} = 0.632$
$2/a$	$1 - e^{-2} = 0.86$
$3/a$	$1 - e^{-3} = 0.95$
$4/a$	$1 - e^{-4} = 0.98$

which is then used to plot the graph of Figure 10.

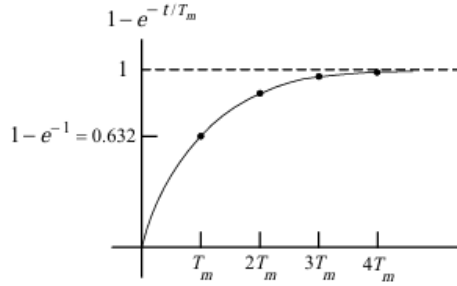


Figure 10: Step response of a first-order system.

In the figure we have set $T_m \triangleq 1/a$ which is called the *time constant* of this first-order system and its units are in seconds. It is a measure of how fast the system responds to an input. The

smaller the value of T_m the faster the system responds. If, for example, v_0 is the desired final angular speed, we would choose

$$V_0 = \frac{v_0}{b/a} = \frac{a}{b}v_0. \quad (11)$$

Then the step input voltage of $V_a(s) = \frac{v_0}{b/a} \frac{1}{s}$ gives a speed response of

$$v(t) = v_0(1 - e^{-at}) = v_0(1 - e^{-t/T_m}) \quad (12)$$

and the smaller the value of T_m the faster $v(t)$ gets to v_0 .

Simulation

- (a) A transfer function model of the cart on the track is given by

$$G(s) \triangleq \frac{b}{s(s+a)}$$

where

$$a = \frac{Rf/r_m^2 + K_b K_T \left(\frac{n_2}{n_1} \frac{1}{r_m} \right)^2}{RJ/r_m^2 + RM}, b = \frac{K_T \frac{n_2}{n_1} \frac{1}{r_m}}{RJ/r_m^2 + RM}. \quad (13)$$

Make a .m file to compute the value of a and b from the motor parameters. Let $M = M_{c2}$. In the backward difference computation of speed, use a sample period of $T = 0.001$ sec. Finally, just set $J_{enc} = 0$ and $f_{enc} = 0$ as they are not given in the parameter table.

Remark The parameter values in this expression are in MKS (meters-kilograms-seconds) units so $X(s)$ will be in meters. Do all of your simulations in MKS. To read $x(t)$ out in centimeters, simply put the output $x(t)$ through a gain block of 100 before putting it into a scope.

- (b) Make a SIMULINK block of the linear cart including a model of the encoder as indicated in Figure 11. To simulate the encoder, one computes number of counts N by finding the greatest integer less than or equal x/K_{enc} or $N = \text{floor}(x/K_{enc})$ as shown in Figure 12. **In all that follows, use this model in any simulation as it captures the effect of the encoder.** Also include a saturation block for the input voltage with $V_{\max} = 5$ V, $V_{\min} = -V_{\max}$.

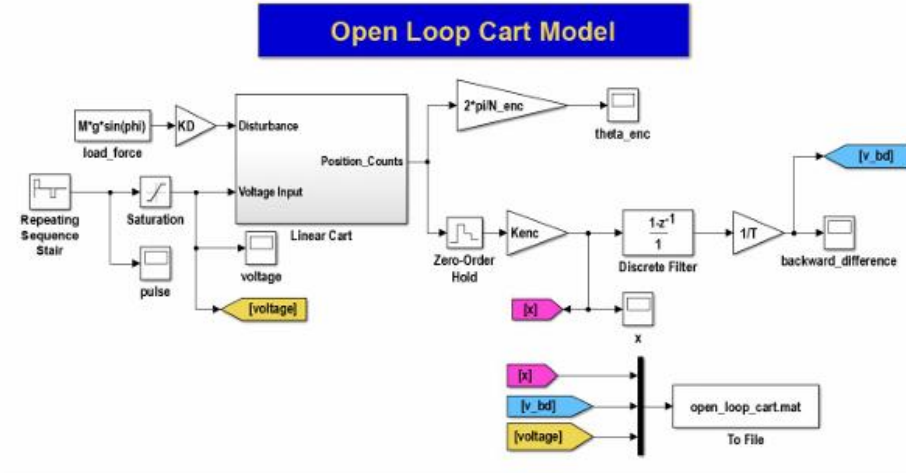


Figure 11: Open loop model of the linear cart.

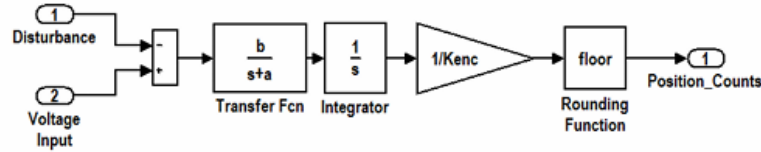


Figure 12: Inside the “Linear Cart” block.

To put the blocks of linear cart model shown in Figure 12, after drawing these blocks, select them, right click, and then choose **create subsystem**.

Let $\phi = 0$, i.e., $F_d = 0$ so there is no load force.

In SIMULINK, click on **simulation/configuration_parameters** to open up the dialog box shown in Figure 13 of Appendix I below. Set the parameter values as shown in the figure.

Similarly, Figures 14, 15, 16 and 17 of Appendix I show how the parameters should be set for the **Zero-Order Hold** block, the **Discrete Filter** block, the **To File** block and the **Repeating Sequence Stair** block, respectively.

Warning: Figure 11 shows two ways to compute the speed from the position measurements. Though both appear to work fine in simulations, in industrial practice the backward difference calculation

$$v_{bd}(kT) \triangleq \frac{x(kT) - x((k-1)T)}{T}$$

is used. **In all that follows, use the backward difference method to obtain the speed by numerical differentiation.** Simulink uses the z transform variable to indicate delays, that is, $z^{-1}\{x(kT)\} \triangleq x((k-1)T)$ and so the **discrete filter** block in Figure 11 implements

$$\frac{1 - z^{-1}}{1} \{x(kT)\} \triangleq \frac{x(kT) - x((k-1)T)}{1}.$$

- (c) Plot the (backward difference) speed and position outputs and verify the error in the speed is bounded by K_{enc}/T . Do your plots using the **plot** command in MATLAB, *not* by using screenshots of your SIMULINK scopes. Further, to do the plotting, write another **.m** file that automatically loads the data and plots everything. To see how to automatically load your data, see "**To File Dialog Block**" in Appendix I below.

Hand in a print out of your MATLAB **.m** file and screen shots of your SIMULINK **.mdl** file.

Make sure your plots look good, that is, the fonts are big enough, the lines in the plots are thick enough, the x and y axis are labeled with appropriate units, and a title is given. Appendix II below gives some examples of using the **plot** command in MATLAB including how to set the **FontSize**, how to do **xlabel** and **ylabel** to label the axes, how to do a **title** for the plot and how to change the **LineWidth** (thickness) of the plots.

- (d) Include in the report your estimate of the system parameters a and b obtained from your simulation plot of the speed versus time. That is, you need to find v_0 and a so that equation (12) approximates your plot of $v(t)$ versus t . The steady-state speed v_0 is found directly from the experimental plot. Next measure the time T_m that the speed $v(T_m) = \frac{dx(T_m)}{dt}$ is equal to $(1 - e^{-1})v_0 = 0.632v_0$. Then

$$T_m = 1/a \text{ or } a = 1/T_m$$

and by (11)

$$b = \frac{v_0}{V_0}a.$$

Compare these estimates of a and b with the values used in your simulation from (13). The point of this simulation work is to show you can obtain the theoretical values of a, b given in (13) from the speed response. Note that your speed measurement is a little noisy which reduces the accuracy in measuring v_0 and T_m and therefore the accuracy in estimating the values of a, b from the speed response.

- e) Repeat part (d) using the data provided

Appendix I:

Simulation/Configuration Parameters Dialog Box

With the SIMULINK software open, go to `simulation/configuration_parameters`. The dialog box shown Figure 13 below will open up. Set Solver options, Type, Solver and fixed-step-size as shown in the Figure 13. Set the sample time to T where $T = 0.001$ is set in your `.m` file.

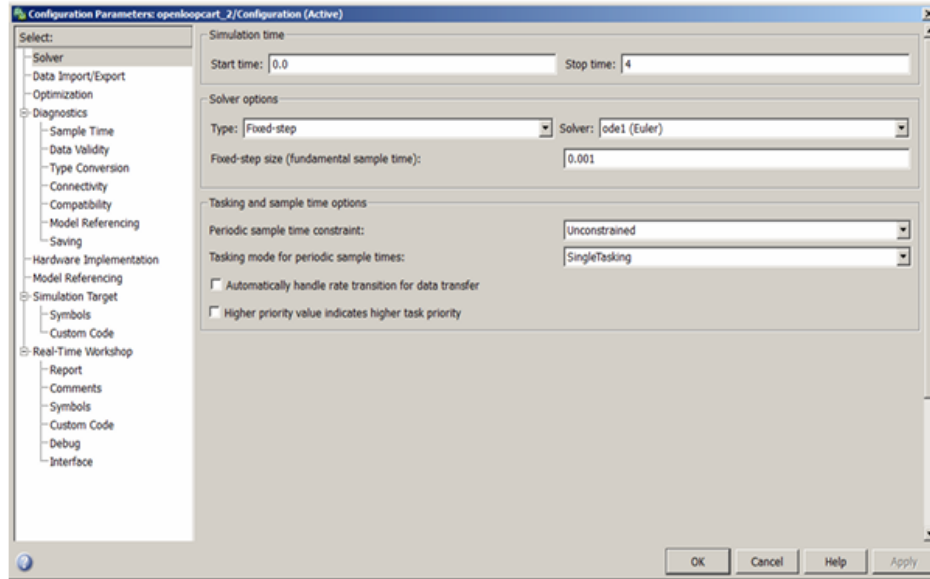


Figure 13: simulation/configuration parameters dialog box.

Zero-Order Hold Dialog Block

Double-click on the Zero-Order Hold block to open up its dialog box which is shown below in Figure 14. Set the sample time to T .

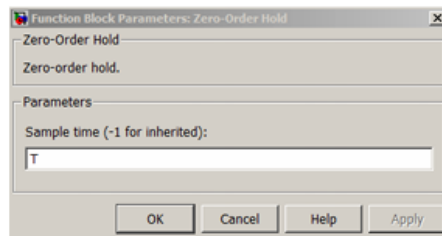


Figure 14: Zero-Order Hold dialog box.

Discrete-Filter Block

Double-click on the **Discrete-Filter** block to open up its dialog box which is shown below. Be sure yours is filled in the same way.

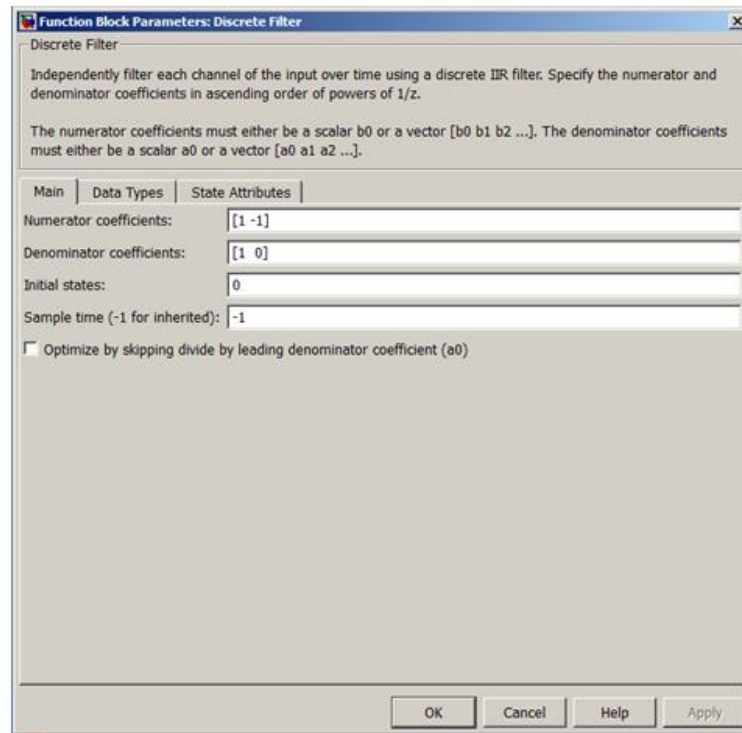


Figure 15: Discrete-Filter dialog box.

To File Dialog Block

Double click on the ToFile block to open up the dialog box below. Fill it out as shown below.

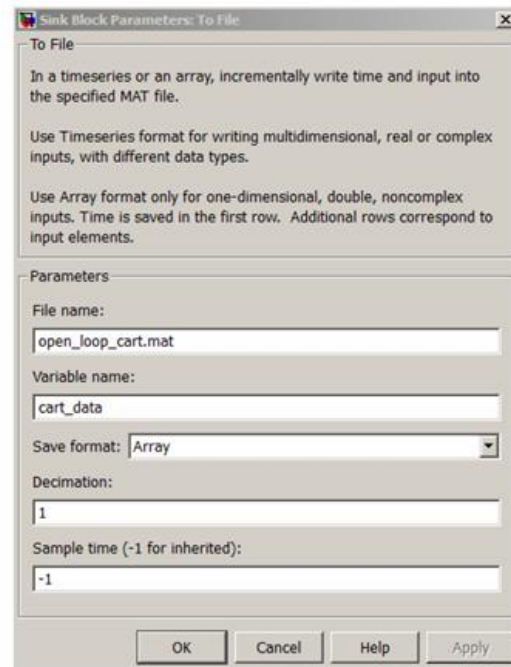


Figure 16: Dialog box for To File block

As seen in Figure 11 there are only three signals that go into the Mux block and then on to the To_File block. However, this block contains four signals because SIMULINK also saves the time variable. This data is stored by SIMULINK in the file `open_loop_cart.mat`. To get the data into the workspace go to the MATLAB window and write the following code:

Repeating Sequence Stair Dialog Block

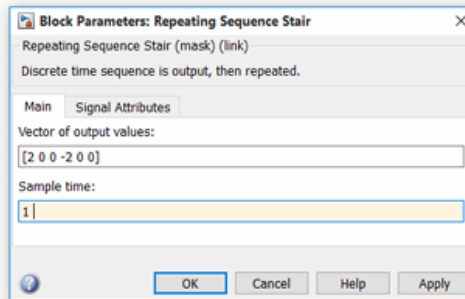


Figure 17: Dialog box for Repeating Sequence Stair.

The Vector of output values given by $[2 \ 0 \ 0 \ -2 \ 0 \ 0]$ specifies the output values of this block. The Sample time given by 1 seconds specifies when to change the output values. So in this example the block puts out 2 for the first 1 secs, 0 for the next 1 secs, 0 for the next 1 secs, -2 for the next 1 secs, 0 for the next 1 secs and finally 0 for the next 1 secs. Then this sequence is repeated, that is, the output of this block is periodic.